



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

SCORE-3.0

11-07-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

Test Type : SCORE-3.0

Test Pattern : JEE (Main)

TEST DATE : 11-07-2021

ANSWER KEY

PAPER-2

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	A	C	D	C	B	C	D	A
SEC-1	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	C	A	B	C	D	C	B	B
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4.00	2.00	0.40	16.00	6.00	500.00	28.00	0.02	80.00	3.00

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	B	B	B	D	A	D	B	C
SEC-1	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	C	D	C	D	A	D	B	D
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	40.00	1.40	8.00	3.00	-122.40	0.25	5.00	270.00	29.00	15.00

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	C	D	B	A	B	C	A	C
SEC-1	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	D	C	C	A	C	D	A	C	A
SEC-2	Q.	1	2	3	4	5	6	7	8	9	10
	A.	14.00	0.00	22.14	80.00	0.60	1.00	1.00	0.50	0.25	0.40 or 0.41

HINT – SHEET

PHYSICS

SEC-1

1. Ans (D)

$$\frac{Q}{i} = \frac{NAB}{C}$$

2. Ans (B)

$$\frac{E}{B} = u$$

$$B = \frac{E}{u} = \frac{\text{kg m/s}^2}{\text{C m/s}} = \text{kg/sec C}$$

3. Ans (A)

$$\therefore \tan \delta = \frac{V}{H}$$

$$\tan 45^\circ = \frac{V}{H \cos 30^\circ}$$

$$(\text{Divide (1) and (2)}) \delta = \tan^{-1} \left(\frac{\sqrt{3}}{2} \right)$$

4. Ans (C)

The coil is combination of two solenoids, connected in series.

If L_a is the coefficient of self-induction with air core and L_r with that for iron core, then

$$L = L_a + L_r$$

$$= \mu_0 n^2 \ell_a + \mu_0 \mu_r n^2 \ell_a$$

$$= \mu_0 n^2 \ell_a (1 + \mu_r) = 1.26 H$$

5. **Ans (D)**

$$V = \frac{LdI}{dt}$$

6. **Ans (C)**

At resonance, $X_L = X_C$

$$Z_{\min} = R$$

7. **Ans (B)**

The direction of B in coil 3 is opposite since windings in the coil in an opposite sense. Since the current coil in 2 and 3 is half that in coil, magnitude of B in and 3 is half that in coil 1. Thus option (B) is correct.

8. **Ans (C)**

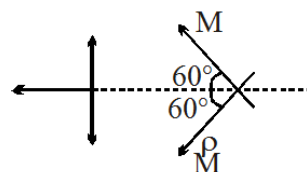
$$\varepsilon = 10 \left(\cos \omega \times \frac{1}{2} + \frac{\sqrt{3}}{2} \sin \omega t \right)$$

$$= 10 \left(\sin \left(\omega t + \frac{\pi}{6} \right) \right)$$

$$i = 5 \sin \left(\omega t + \frac{\pi}{4} \right)$$

current leads voltage $\Rightarrow$ circuit is capacitive

9. **Ans (D)**



$$B = \frac{2\mu_0 M}{4\pi d^3} \times \cos 60^\circ \times 2$$

10. **Ans (A)**

$$\omega = \frac{\text{angular momentum}}{mR^2(M.I.)} = \frac{J}{mR^2}$$

Ring is equivalent to a current loop of current

$$I = \frac{Q}{2\pi} \omega$$

magnetic moment of loop

$$\vec{\mu} = nI\vec{A} = (1) \left(\frac{Q}{2\pi} \omega \right) \pi R^2 \hat{n} = \left(\frac{Q}{2} \omega R^2 \right) \hat{n}$$

($\hat{n}$ is normal to plane outward $\odot$)

$$\text{Torque } \vec{\tau} = \vec{\mu} \times \vec{B}$$

$$= \frac{Q}{2} \omega R^2 B (\text{up the plane})$$

$$= \frac{Q}{2} R^2 \left(\frac{J}{mR^2} \right) B = \frac{1}{2} \left(\frac{Q}{m} \right) (J) (B)$$

$$= \frac{1}{2} \times 10^{-5} \times 7.5 \times 10^{-4} \times 0.1$$

$$= 3.75 \times 10^{-10} \text{ N upward along the plane}$$

11. **Ans (D)**

Lenz's law

12. **Ans (B)**

$$v = \frac{dx}{dt} = 2 \times 10^{-2} \times 120\pi \sin(120\pi t)$$

$$= 2.4 \pi \sin 120\pi t$$

$$r = B\ell v = 1.2 \times 0.15 \times 2.4\pi \sin 120\pi t$$

$$= 0.432 \pi \sin 120\pi t$$

13. **Ans (C)**

$$\text{COP} = \frac{Q_{\text{extracted}}}{\text{work done}} = \frac{T_1}{T_2 - T_1} = \frac{270}{30} = 9$$

$$\Rightarrow \text{work} = \frac{1000}{9} \text{ J}$$

14. **Ans (A)**

$$207 = C_p \times 10$$

$$\Delta Q = C_v \times 5$$

$$C_v = C_p - R = 12.4$$

$$\Delta Q = nQ\Delta T = 62 \text{ J}$$

$$C_p = 207$$

$$C_v = 20.7 - 8.3 = 12.4$$

$$\Delta Q = 1 \times 12.4 \times 5$$

16. **Ans (C)**

Number of moles present initially is $n = \frac{PV}{RT}$.

Let n' be the number of moles of the gas that leaked till the pressure falls to P' . Since volume V of the vessel cannot change and temperature T remains constant during leakage, we have

$$n' = \frac{P'V}{RT}$$

$\therefore$ Number of moles that leaked is

$$\Delta n = n - n' = \frac{PV}{RT} - \frac{P'V}{RT}$$

$$= \frac{V}{RT} (P - P')$$

So the correct choice is (C)

17. **Ans (D)**

CA is adiabatic, BC is isothermal as per slope.

$$\text{Hence } V_1 = \frac{P_0 V_0}{3P_0} = \frac{V_0}{3}$$

$$\text{using } PV^r = \text{const.} \Rightarrow r = \frac{\ln 6}{\ln 3}$$

18. **Ans (C)**

$$PV = \frac{m}{M} RT$$

$$\frac{m}{M} R(T + \Delta T)$$

$$50(300 + \Delta T) = 100(300)$$

$$\Delta T = 300 \text{ K}$$

19. **Ans (B)**

$$K_{av} = \frac{5}{2} kT \{H_2\}$$

$$K_{av} = \frac{3}{2} kT \{Ar\}$$

$$p_1 = \frac{n_1}{n_1 + n_2} \times p$$

$$n_1 > n_2 \Rightarrow p_1 > p_2$$

20. **Ans (B)**

$$\frac{dH}{dt} = ms \frac{dT}{dt} = \text{constant}$$

$$\text{slope of graph } \left(\frac{dT}{dt} \right) =$$

constant

$$\text{mass} \times \text{specific heat}$$

PHYSICS

SEC-2

1. **Ans (4.00)**

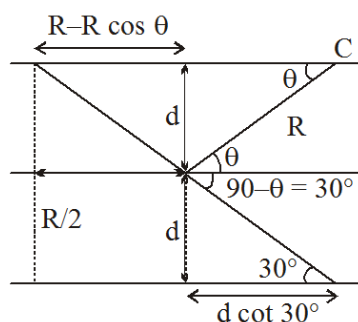
$$F = 12\hat{i} - 8\hat{j} = q(\vec{v} \times \vec{B})$$

$$= 2\hat{i} - 3\hat{j} \times B_0\hat{k}$$

$$= -2B_0\hat{j} = -3B_0\hat{i}$$

$$B_0 = 4T$$

2. **Ans (2.00)**



$$\sin \theta = \frac{d}{R} = \frac{\sqrt{3}}{2}$$

$$\theta = 60^\circ$$

$$x = \frac{R}{2} + \frac{R\sqrt{3}}{2} \times \sqrt{3} = 2R$$

3. **Ans (0.40)**

$$B = \frac{\mu_0 Ni}{2\pi R}$$

$$H = \frac{B}{\mu_0} = \frac{Ni}{2\pi R}$$

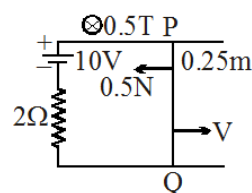
$$B = \mu_0 (H + M)$$

$$M = x H$$

$$\frac{B}{\mu_0 H} \times 100 = 1 + x = (1 + 4 \times 10^{-3}) \times 100$$

4. **Ans (16.00)**

$$\text{Induced emf} = B\ell v = 0.125 \text{ V}$$



$$\text{current } I = \frac{10 - e}{R}$$

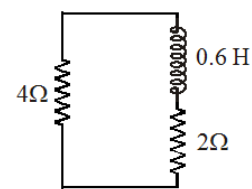
$$\text{Force } BIl = 0.5 \left(\frac{0.125 - 10}{2} \right) 0.25$$

$$= 0.5 \text{ N (given)}$$

$$\text{Solving } v = 16 \text{ m/s}$$

5. **Ans (6.00)**

$$i_2 = 6A$$



$$t = 6e^{-\frac{6t}{0.6}} = 6e^{-10t}$$

6. **Ans (500.00)**

$$\frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \omega L - \frac{1}{\omega C} = \pm 10\Omega \Rightarrow 2\omega L = \frac{1}{\omega C}$$

$$20 = \frac{1}{100C}$$

$$C = \frac{1}{2000} \text{ F} = 500 \mu\text{F}$$

7. **Ans (28.00)**

$$B = \frac{\mu_0 I}{2\pi} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$= 2 \times 10^{-7} \times 5.6 \times \frac{1}{0.4}$$

$$= 28 \times 10^{-7} \text{ T}$$

8. **Ans (0.02)**

$$\vec{B} = \frac{1}{\pi} \left(\frac{Wb}{m^2} \right)$$

Area of the disc normal to $\vec{B}$ is $\pi R^2 \cos 60^\circ$

Flux = $B \times$ Area normal

$$\therefore \text{Flux} = \frac{1}{2} \times .04 = 0.02 \text{ Wb}$$

9. **Ans (80.00)**

$$\frac{3}{4} \text{ mL} = Q$$

$$Q = \frac{\text{mL}}{4} + mS \times 40$$

$$\frac{\text{mL}}{2} = 4mS$$

$$\frac{L}{S} = 80$$

10. **Ans (3.00)**

$$m \times 80 = 100 \times 1 \times (20 - 0)$$

$$m = 25$$

$$n = \frac{25}{10} = 2.5$$

CHEMISTRY

SEC-1

1. **Ans (D)**

Ferrous ion gives white ppt with $K_4[Fe(CN)_6]$

2. **Ans (B)**

dil. HCl is first group reagent so it can easily separate Hg^{2+} & Ag^+ . Because Hg^{2+} is second group cation.

4. **Ans (B)**

$[Fe_3(OH)_2(CH_3COO)_6]^+$ is formed.

5. **Ans (B)**

$HgCl_2$ is very weakly ionised in aqueous medium due to more covalent character.

10. **Ans (C)**

From copper pyrites, Iron sulphide is removed by its oxidation then slag formation.

11. **Ans (D)**

$Fe^{2+}_{(aq)}$ gives above reactions

15. **Ans (C)**

Commercially carbon reduction is used for extraction of Zinc.

16. **Ans (D)**

Pure aluminium floats above molten electrolyte during refining of 'Al'.

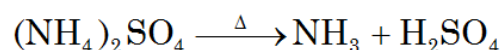
CHEMISTRY

SEC-2

9. **Ans (29.00)**

Malachite $[CuCO_3 \cdot Cu(OH)_2]$ is ore of copper (atomic number $Z = 29$).

10. **Ans (15.00)**



15 atoms

MATHEMATICS

SEC-1

1. **Ans (D)**

$$p \quad q \quad [(\sim (p \wedge p) \rightarrow q)] \rightarrow (p \rightarrow p)$$

$$T \quad T \quad T$$

$$T \quad F \quad T$$

$$F \quad T \quad T$$

$$F \quad F \quad T$$

2. **Ans (B)**

$${}^{10}C_r \cdot (\sqrt{x})^{10-r} \cdot \left(-\frac{k}{x^2} \right)^r$$

$$(-1)^r \cdot {}^{10}C_r \cdot x^{\frac{10-r}{2} - 2r} k^r$$

$$\frac{10-r}{2} - 2r = 0$$

$$r = 2$$

3. **Ans (C)**

$p \vee \sim p$ is a tautology

4. **Ans (D)**

$$r(20-r) \cdot {}^{20}C_r \cdot {}^{20}C_r$$

$$20 \times 19 \cdot {}^{18}C_{r-1} \cdot {}^{20}C_r$$

$$20 \times 19 \sum_{r=1}^{r=19} {}^{18}C_{r-1} \cdot {}^{20}C_{20-r}$$

$$20 \times 19 \cdot {}^{38}C_{19} = 20 \times 19 \times \frac{38!}{19!19!}$$

$$= 20^2 \times \frac{38!}{18!20!}$$

$$= 400 \cdot {}^{38}C_{20}$$

5. **Ans (B)**

$$(2,2) \notin R$$

$$(3,6) \in R \Rightarrow (6,3) \in R$$

$$(6,15) \in R, (15,18) \in R \Rightarrow (6,18) \notin R$$

6. **Ans (A)**

coefficient of x^4

$${}^{15}C_4 + {}^{16}C_4 + {}^{17}C_4 + \dots + {}^{15}C_4$$

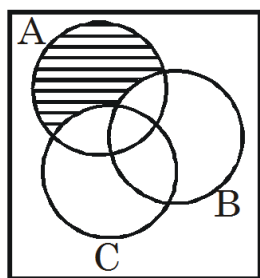
$${}^{15}C_5 + {}^{15}C_4 + {}^{16}C_4 + \dots + {}^{15}C_4 - {}^{15}C_5$$

$${}^{16}C_5 + {}^{16}C_4 + \dots + {}^{15}C_4 - {}^{15}C_5$$

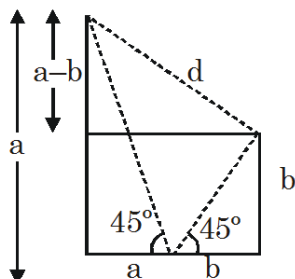
$${}^{17}C_5 + \dots + {}^{50}C_4 - {}^{15}C_5$$

$$\Rightarrow {}^{51}C_5 - {}^{15}C_5$$

7. **Ans (B)**



8. **Ans (C)**



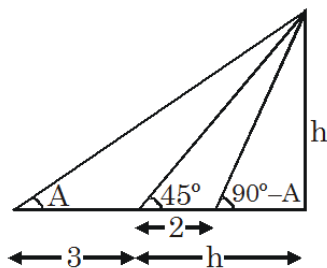
$$d = \sqrt{(a-b)^2 + (a+b)^2} = \sqrt{2(a^2 + b^2)}$$

9. **Ans (A)**

$$(x_1 + x_2 + x_3 < 20) - (x_1 + x_2 + x_3 < 15)$$

$${}^{20}C_3 - {}^{12}C_3$$

10. **Ans (C)**



$$\tan A = \frac{h}{3+h} \quad \dots(1)$$

$$\tan(90 - A) = \frac{h}{h-2} \quad \dots(2)$$

multiply (1) and (2)

$$1 = \frac{h^2}{h^2 + h - 6}$$

$$h = 6m$$

11. **Ans (D)**

$$\underbrace{{}^5C_3}_{\text{selection of girls}} \times \underbrace{2^4}_{\text{selection of boys}}$$

12. **Ans (D)**

$${}^5C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3 \cdot \left(\frac{1}{6}\right) \times 1^4$$

13. **Ans (C)**

$$\frac{4!}{2!} \frac{1}{2!} \times \frac{5!}{2!3!}$$

14. **Ans (C)**

$$\begin{aligned} &= i + i + i^2 + i^6 + i^{24} + i^{120} + \dots \\ &= 2i - 1 - 1 + 97 \\ &= 95 + 2i \end{aligned}$$

15. **Ans (A)**

abc

$b \neq 0, 1$ possible cases of $b = 2, 3, \dots, 9$

number of three digit numbers

$$= 1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + 8 \times 9$$

16. Ans (C)

$$\begin{aligned} & \arg \left(\frac{(1+i)^5 (1+\sqrt{3}i)^2}{(-2i)(-\sqrt{3}+i)} \right) \\ &= 5\arg(1+i) + 2\arg(1+\sqrt{3}i) \\ & \quad - \arg(-2i) - \arg(-\sqrt{3}+i) \\ &= \frac{5\pi}{4} + \frac{2\pi}{3} + \frac{\pi}{2} - \frac{5\pi}{6} = \frac{19\pi}{12} \\ &= \left(2\pi - \frac{19\pi}{12} \right) = -\frac{5\pi}{12} \end{aligned}$$

17. Ans (D)

$$\begin{aligned} & \left| z - \frac{4}{z} \right| = 2 \\ & \left| z - \frac{4}{z} + \frac{4}{z} \right| \leq \left| z - \frac{4}{z} \right| + \frac{4}{|z|} \\ & |z| \leq 2 + \frac{4}{|z|} \\ & \Rightarrow |z| \leq 1 + \sqrt{5} \end{aligned}$$

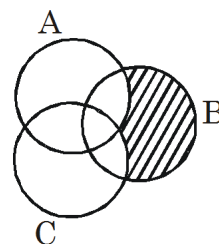
18. Ans (A)

$$\begin{aligned} & a^4b^4 + c^4 - 2a^2b^2 - 2b^2c^2 + 2a^2c^2 = 2a^2c^2 \\ & (c^2 + a^2 - b^2)^2 = 2a^2c^2 \\ & \frac{c^2 + a^2 - b^2}{2ca} = \pm \frac{1}{\sqrt{2}} \\ & \cos B = \pm \frac{1}{\sqrt{2}} \end{aligned}$$

19. Ans (C)

$$\begin{aligned} & 2b^2 \sin 2C \cos C + 2C^2 \sin B \cos B = 2bc \\ & 2b^2 \frac{c}{2R} \cos C + 2c^2 \frac{b}{2R} \cos B = 2bc \\ & \frac{bc}{R} (b \cos C + c \cos B) = 2bc \\ & a = 2R \\ & \sin A = 1 \\ & A = 90^\circ \\ & \Delta = \frac{1}{2} bc \sin A = 210 \end{aligned}$$

20. Ans (A)



$$\begin{aligned} & P(A \cap B \cap C) = 1 - P(\overline{A \cap B \cap C}) \\ &= 1 - \frac{15}{16} = \frac{1}{16} \\ & P(A^c \cap B \cap C^c) \\ &= P(B) - (P(A \cap B) + P(B \cap C) - P(A \cap B \cap C)) \\ &= \frac{3}{4} - \left(\frac{1}{8} + \frac{1}{4} - 1 \times \frac{1}{16} \right) \end{aligned}$$

MATHEMATICS

SEC-2

1. Ans (14.00)

S.D. is independent of origin

$$\begin{aligned} & \sigma_{\text{new}} = \frac{2}{7} \sigma_{\text{old}} \\ &= \frac{2}{7} \times 49 = 14 \end{aligned}$$

2. Ans (0.00)

$$\begin{aligned} & (1+x)((1+x)(1-x+x^2))^{100} \\ & (1+x)(1-x^3)^{100} \\ & (1-x^3)^{100} + x(1-x^3)^{100} \\ & \text{coeff of } x^{50} \\ &= 0 + 0 = 0 \end{aligned}$$

3. Ans (22.14)

Use formula

$$\sigma^2 = \frac{n_1 \sigma_1^2 + n_2 \sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2} (\bar{x}_1 - \bar{x}_2)^2$$

4. Ans (80.00)

coeff. of x on both side

$$a_1 = -{}^b c_1 + {}^a c_1 \quad \dots (1)$$

coeff. of x^2 on both side

$$a_2 = {}^a c_2 - {}^a c_1 {}^b c_1 + {}^b c_2 \quad \dots (2)$$

5. **Ans (0.60)**

$$a + b = 4c$$

$$a + b + c = 5c$$

$$25 = 5c$$

$$\tan \frac{A}{2} \tan \frac{B}{2} = \frac{\Delta}{s(s-a)} \frac{\Delta}{s(s-b)}$$

$$= \frac{s-c}{s} = \frac{\frac{5c}{2} - c}{\frac{5c}{2}} = \frac{3}{5}$$

6. **Ans (1.00)**

$$\left| \frac{\frac{z_1}{z_2} - 1}{\frac{z_1}{z_2} + 1} \right| = \left| \frac{\lambda i - 1}{\lambda i + 1} \right|^4 = 1$$

7. **Ans (1.00)**

$$|a| = 1 \Rightarrow a\bar{a} = 1$$

$$|b| = 1 \Rightarrow b\bar{b} = 1$$

$$|c| = 1 \Rightarrow c\bar{c} = 1$$

$$a + b + c = abc$$

$$\Rightarrow \frac{1}{ab} + \frac{1}{bc} + \frac{1}{ca} = 1$$

$$\bar{a}\bar{b} + \bar{b}\bar{c} + \bar{c}\bar{a} = 1$$

$$\Rightarrow ab + bc + ca = 1$$

$$\Rightarrow |ab + bc + ca| = 1$$

8. **Ans (0.50)**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$.7 = .4 + p - .4 \times p$$

$$p = \frac{1}{2}$$

9. **Ans (0.25)**

$$3^{45} \cdot 5^{45}$$

$$\text{Number of divisors} = 46 \times 46$$

$$15^{22} = 3^{22} \cdot 5^{22}$$

$$\text{Number of divisors} = 23 \times 23$$

$$P = \frac{23 \times 23}{46 \times 46} = \frac{1}{4}$$

10. **Ans (0.40 or 0.41)**

$$P = \frac{(1 + 5 + 6 + 1)}{2^5} = \frac{13}{32}$$